FLOWER WELFARE

A first check of flower condition

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Hackathon 7 · Eyes on Earth · SDG 15, Life on Land

PROBLEM

- **181 of the 331** Dutch wild bee species (55%) are on the Red List [1].
- In the EU, **1 in 3** bee, butterfly and hoverfly species is disappearing [2].
- Pollinators need flowers in good condition to feed.
- Den Haag is adding flower meadows [7], but did not monitor green maintenance quality before 2024 [8].

[1] Reemer 2018, Rode Lijst Bijen (natuurtijdschriften.nl)
[2] European Commission, 24 Jan 2023
[7] Gemeente Den Haag, Stadsnatuur
[8] Rekenkamer Den Haag 2025, Koers op Groen

SOLUTION

- Upload one photo of a bed; our Roboflow model marks each flower **Healthy** or **Wilted**.
- In the flowering season, the Healthy share shows where to water, delay mowing or reseed first.
- **Why not just visit?** One ecologist can triage many beds from photos before going out.
- **Not for:** private gardens, plant disease diagnosis or automatic mowing decisions.

SDG 15 & OUR USER

SDG 15.5

“Take urgent and significant action to reduce the degradation of natural habitats, halt the loss of biodiversity…” [3]

USER

A green-space ecologist at the **Gemeente Den Haag**, which is adding flower meadows and wants nature-friendly mowing [7]. Decides where to water, delay mowing or reseed first.

WHY

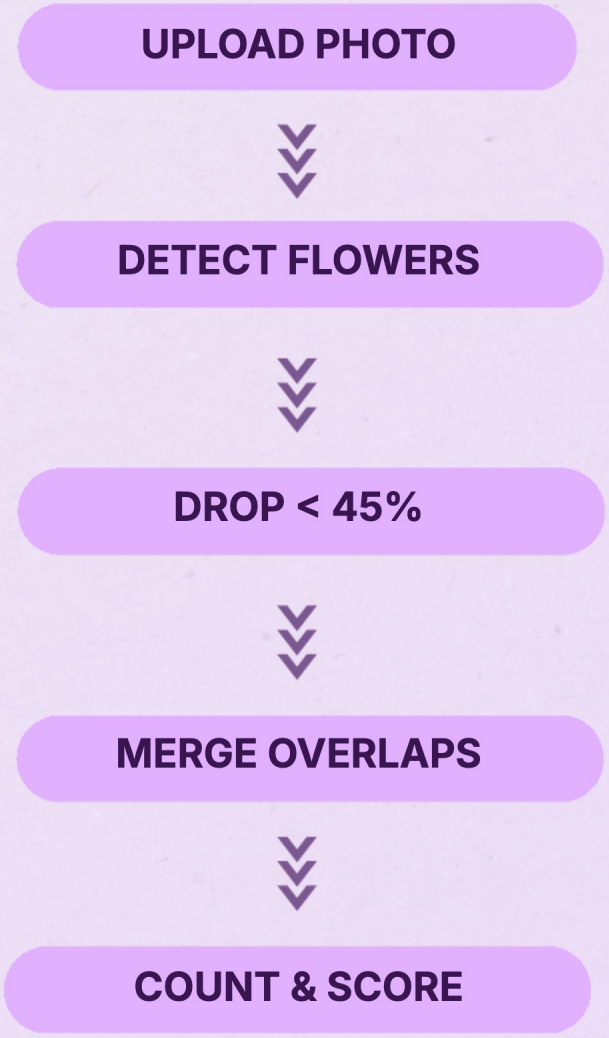
Den Haag did not monitor the quality of its green maintenance before 2024 [8]. About 60% of Dutch municipalities now do ecological verge management [9].

[3] United Nations, SDG target 15.5 · [7] Gemeente Den Haag, Stadsnatuur
[8] Rekenkamer Den Haag 2025, Koers op Groen
[9] VNG 2025, factsheet ecologisch bermbeheer (citing Vlinderstichting 2024)

HOW IT WORKS

The app sends a photo to our Roboflow model. It draws a box around each flower it finds and labels it Healthy or Wilted.

Boxes below 45% confidence are dropped, overlapping boxes are merged, and the app shows the counts and the share of Healthy flowers.



HOW WE BUILT IT

Data. 317 photos [5]: 197 from iNaturalist, filtered to the Netherlands, and 120 from Wikimedia Commons (dead, wilting and dried flower categories).

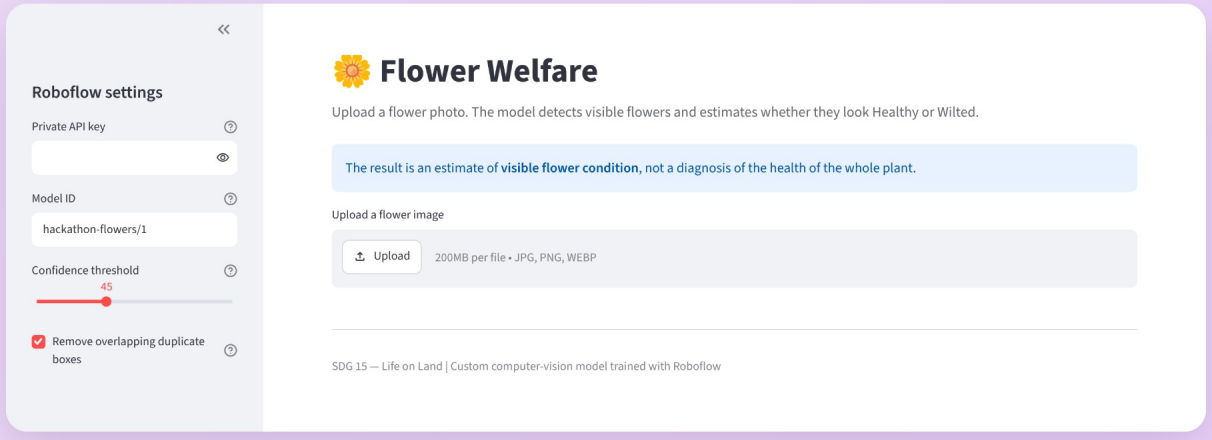
Labels. We labelled the first 30 photos by hand. Roboflow auto-labelled the rest, and then we approved, changed or discarded every auto-label.

Split. 224 train · 65 valid · 28 test photos; the training photos were augmented to 432 images [5].

Model. Roboflow RF-DETR Object Detection, trained on two classes: Healthy and Wilted [6].

[5] Roboflow project hackathon-flowers v1 (dataset export and metrics) · [6] RF-DETR documentation (Roboflow)

LIVE DEMO



The Flower Welfare app (Streamlit), before a photo is uploaded.

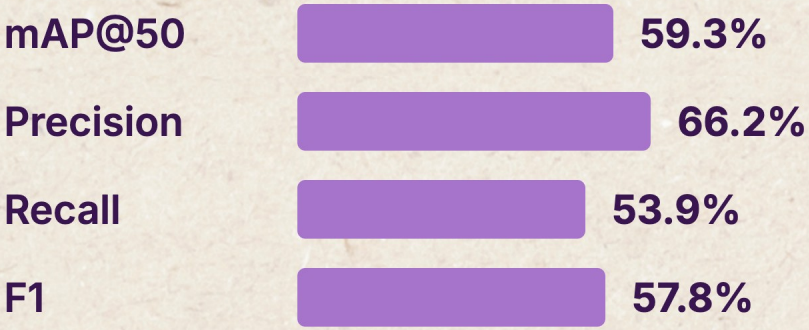
1. Upload a photo the model has never seen.

2. Press Analyze flower image.

3. Green box = Healthy, red box = Wilted, plus the counts and the Healthy share.

App link:

RESULTS



It misses almost half of the flowers, and about 1 in 3 boxes is wrong. Measured on only 28 test images, so these numbers are rough.



Dense field: only a few dozen flowers boxed, most missed.



Close-up: open flowers and seed heads found, buds ignored.

[5] Roboflow project hackathon-flowers v1, test-set metrics (28 images) · failure examples from our own tests

ETHICS & REFLECTION

DATA BIAS	90% of Healthy boxes are in iNaturalist photos and 66% of Wilted boxes in Commons photos [5], so the model may partly learn photo style.
MISSED FLOWERS	Recall is 53.9% on 28 test images, which hold 254 Healthy vs 69 Wilted boxes [5]. Dense scenes are missed most.
WILTED ≠ BAD	Seed heads and dry stems are part of a healthy habitat. Never use the output to remove flowers.
LICENCES	iNaturalist photos default to CC BY-NC [4]: credit the photographers, no commercial use.
OUR TAKE	A first screen that shows where to look, never a decision on its own. Next: our own photos for both classes and a bigger test set.

[4] iNaturalist Help, licences · [5] Roboflow project hackathon-flowers v1

THANK YOU

Questions?

Sources

[1] Reemer 2018, Rode Lijst Bijen (review) · [2] European Commission 2023
[3] UN SDG 15.5 · [4] iNaturalist licences · [5] Roboflow project hackathon-flowers v1
[6] RF-DETR docs · [7] Gemeente Den Haag, Stadsnatuur · [8] Rekenkamer Den Haag 2025 · [9] VNG 2025